Daniel Larson

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Education

University of California, Santa Barbara — Santa Barbara, CA
B.S. in Statistics and Data Science | GPA: 3.50 — *Sep. 2024 – Dec. 2026*
- Relevant Courses: Principles of Data Science, Regression Analysis, Applied Stochastic Processes, Statistical Learning, Design and Analysis of Experiments, Time Series, Big Data Analysis, STAT Data Science.

College of the Canyons — Santa Clarita, CA
Completed IGETC certified coursework | General Chemistry I & II, Statistics, Calculus I & II — *Aug. 2021 – Jun. 2024*

Experience

Tenart Tennis Platform Operations Intern — Jun. 2026 – Aug. 2026
Shanghai Buba Sports & Culture Co., Ltd. (Tenart) — *Shanghai, China*
- Built and maintained SQLite database pipeline integrating Feishu API with platform trend logging; implemented validation framework ensuring 100% data integrity.
- Engineered a Python web scraper (Selenium, undetected-chromedriver, BeautifulSoup) and SQLite-backed competitive intelligence system processing 500+ reviews across 162 Shanghai tennis venues; surfaced quantified gaps (e.g., parking friction cited 324 times across 40 venues) that informed Tenart's listing optimization strategy.
- Applied Chinese NLP (jieba tokenization, stopword filtering, TF-IDF vectorization) and K-Means clustering to segment competitor venues by review theme, validating cluster stability via bootstrap resampling (Adjusted Rand Index).
- Conducted platform algorithm verification research across Douyin, Xiaohongshu, and Dianping; published bilingual playbooks synthesizing 27 primary sources with confidence-tiered findings, replacing unvetted AI research with validated industry consensus.
- Optimized search visibility through systematic keyword research (85-term matrix across 5 customer personas and 3 platforms) and built an ARIMA time-series forecasting model (ADF stationarity testing, ACF/PACF order selection, AIC-based model comparison) on 35+ weeks of Baidu Index data to project tennis search demand trends.
- Delivered weekly bilingual reports (Chinese/English) to founder synthesizing trend analysis, competitive benchmarks, and KPI recommendations; operated across on-court coaching and data science functions in fast-paced startup environment.

Projects

SpiceRack – *Python, Flask, Scikit-learn, NLP, TF-IDF, K-Means, SQLite, REST APIs* — Oct. 2025 – May. 2026
- Led a team as project manager to architect and ship a recipe recommendation web app, delegating tasks across ML, backend, and frontend tracks to meet a fixed showcase deadline.
- Designed a custom NLP pipeline (TF-IDF with IDF^2 boosting → TruncatedSVD → MiniBatchKMeans) trained on 2.2M+ recipes, resolving cluster dominance and sparse-input failure modes to achieve a silhouette score of 0.78 across ~475 clusters.
- Engineered a top-K cosine similarity search with thread-safe REST API image caching and a SQLite-backed saved recipes system, reducing recommendation latency and enabling seamless AJAX interactions.
- Selected as 3rd place finalist out of 100+ teams at the UCSB Data Science Club Project Showcase, validating end-to-end ML deployment from raw data to production web app.

MLB Pitcher Injury Risk Prediction – *R, XGBoost, K-Means, tidymodels* — Jan. 2026 – Mar. 2026
- Analyzed 30 years of MLB pitching data from the Lahman Database to predict pitcher injury risk in the following season using supervised and unsupervised ML methods.
- Built and compared 4 machine learning models (logistic regression, elastic net, random forest, XGBoost), identifying total innings pitched – including postseason – as the leading risk factor.
- Applied K-means clustering to segment pitchers into 3 workload archetypes; the highest-workload group showed nearly double the injury rate (26.6%) compared to the lowest (14.1%).
- Optimized the final model to flag 80% of at-risk pitchers correctly, with the top 10% highest-risk group injured at a 53.6% rate – nearly 3x the 19% baseline average.

Activities

Data Science Club (UCSB) — Oct. 2024 – Present
- Won "Best Sports Analyst Project" at Datathon 2025 and was selected as a 2-time Project Pitchfire Finalist (SpiceRack, Anti-DUI) among competing student teams.

Skills

Programming Languages: Python, R, SQL, C++
Concepts: Machine Learning, Statistical Modeling, Classification, Clustering, Regression, Natural Language Processing, Tokenization, Time Series Forecasting, Computer Vision, Feature Engineering, A/B Testing, REST APIs
Libraries & Frameworks: Scikit-Learn, XGBoost, tidymodels, statsmodels, Flask, Pandas, NumPy, OpenCV, Selenium, BeautifulSoup, jieba
Tools: Microsoft Excel, Git, GitHub, Jupyter Notebook, VS Code, Docker, SQLite, Feishu API
Languages (Spoken): English (Native), Spanish (Elementary), Mandarin Chinese (Beginner)
Achievements: Datathon 2025 – Best Sports Analyst Project, 2x Project Pitchfire Finalist (SpiceRack, Anti-DUI), Dean's Honors (LS)